Generative Market Simulation

Reproducing Stylized Facts with Deep Generative Models

EECS 4904 — Spring 2026 Final Project

Shufeng Chen · Yixuan Ye · Yizheng Lin · Kevin Sun · Yuxia Meng

Problem Statement

Why Synthetic Market Data?

- Risk management demands thousands of realistic scenarios to stress-test portfolios — replaying history yields only one realized path
- Classical models (GARCH) produce conditionally Gaussian returns, failing to capture heavy tails, volatility clustering, and leverage effects
- **Goal:** Train deep generative models to produce synthetic multi-asset return series that faithfully reproduce real financial data's statistical properties

Dataset

16 Assets

S&P 500 sector ETFs, Treasuries, gold, oil

2005–2026

Daily returns, ~21 years of market history

~5,300 Windows

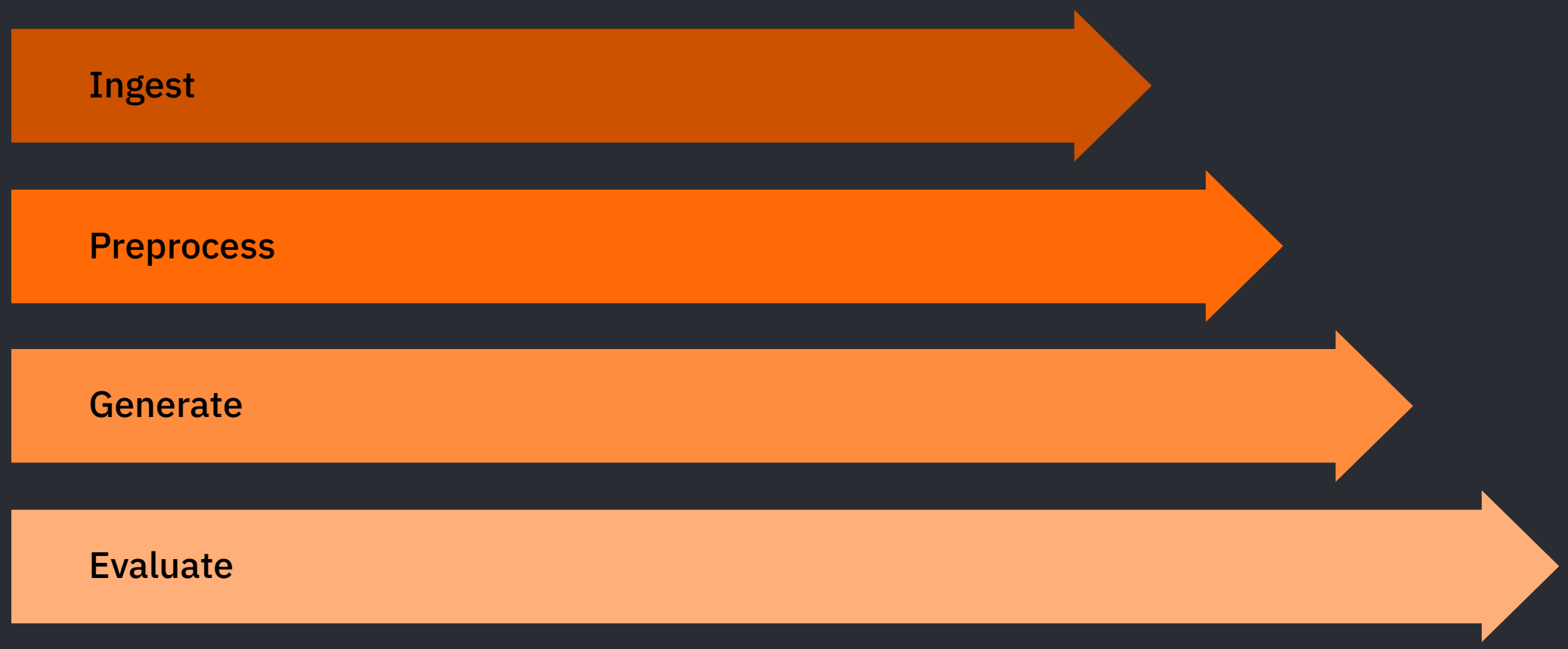
60-day overlapping windows, stride = 1

Six Stylized Facts of Financial Returns

Every generated dataset is validated against six well-documented empirical regularities of financial markets.

1	**Fat Tails** Heavier-tailed than Gaussian — Hill estimator **$\alpha < 5$**
2	**Volatility Clustering** Large moves follow large moves — GARCH(1,1) **$\gamma > 0.85$**
3	**Leverage Effect** Negative returns amplify subsequent volatility — GJR-GARCH **$\gamma > 0$**
4	**Long Memory** Slow ACF decay in |returns| — Hurst exponent **$0.5 < H < 1.0$**
5	**Cross-Asset Correlations** Time-varying inter-asset correlations — max eigenvalue **$\lambda_1 > 1.5$**
6	**No Raw Autocorrelation** Raw returns are approximately i.i.d. — **MAA < 0.05** + Ljung-Box

Architecture Overview



Five models span the full spectrum from classical statistical baselines to state-of-the-art diffusion, enabling rigorous head-to-head comparison under a unified evaluation framework.

1 **GARCH**
Statistical baseline

2 **VAE**
BiGRU encoder-decoder

3 **TimeGAN**
Adversarial + supervisor

4 **RealNVP**
Normalizing flow

5 **DDPM**
1-D U-Net diffusion

BASELINE

GARCH(1,1) — Classical Baseline

Model Design

- Per-asset GARCH(1,1) with AR(1) mean and Student-t innovations
- Cross-asset dependence via Cholesky decomposition of empirical residual correlation matrix
- Role: establishes a non-deep-learning reference point

Results (3-seed avg)

SF: 1.3 / 6

Only leverage effect (1/3 seeds)

MMD: 0.042

Moderate sample quality

Disc: 1.00

Trivially distinguishable from real data

⚠ Fails Fat Tails, Volatility Clustering, Long Memory, Cross-Asset Correlations, and No Raw Autocorrelation

DEEP MODEL

VAE — Bidirectional GRU

Model Design

- Bidirectional GRU encoder with autoregressive decoder
- Student-t likelihood with learnable degrees of freedom
- Factor loadings for cross-asset structure; ACF and correlation regularizers

Results (3-seed avg)

SF: 1.0 / 6

Leverage effect only

MMD: 0.020

Competitive sample quality

Disc: 0.75

Partially deceives discriminator

⚠ Low MMD but struggles with distributional structure — most stylized facts fail despite good sample-level quality

DEEP MODEL

TimeGAN — Adversarial Temporal Model

Model Design

- Embedding, recovery, generator, supervisor, and discriminator networks
- Trained with WGAN-GP losses and moment matching for temporal coherence
- Designed specifically for sequential data fidelity

Results (3-seed avg)

SF: 4.0 / 6

Fat tails, clustering, leverage, long memory ✓

MMD: 0.110

Highest MMD among deep models

Disc: 1.00

Trivially distinguishable

ⓘ Strong temporal dynamics — fails cross-asset correlations and raw autocorrelation; trades distributional quality for temporal realism

DEEP MODEL

Normalizing Flow — RealNVP

Model Design

- RealNVP with affine coupling layers and ActNorm for stable training
- Exact log-likelihood training on flat-vector window representations
- Invertible architecture enables exact density estimation

Results (3-seed avg)

SF: 5.0 / 6

All except No Raw Autocorrelation

MMD: 0.027

Good distributional fidelity

Disc: 0.73

Best discriminative score — closest to ideal 0.5

Synthetic data hardest to distinguish from real across all non-DDPM models — tied with DDPM on SF count

BEST MODEL

DDPM — Our Best Model

Key Innovations

- **V-prediction objective** (Salimans & Ho, 2022) — single most impactful change, SF: 1.7/6 → 5.0/6
- **Student-t forward process** (df=5) — 6× MMD reduction (0.037 → 0.006)
- Cosine noise schedule, DDIM sampling, EMA, classifier-free guidance
- **Key discovery:** Sigmoid schedule suppresses volatility clustering with v-prediction (SF: 5.0 → 2.7); cosine is the correct pairing

Results (3-seed avg)

SF: 5.0 / 6

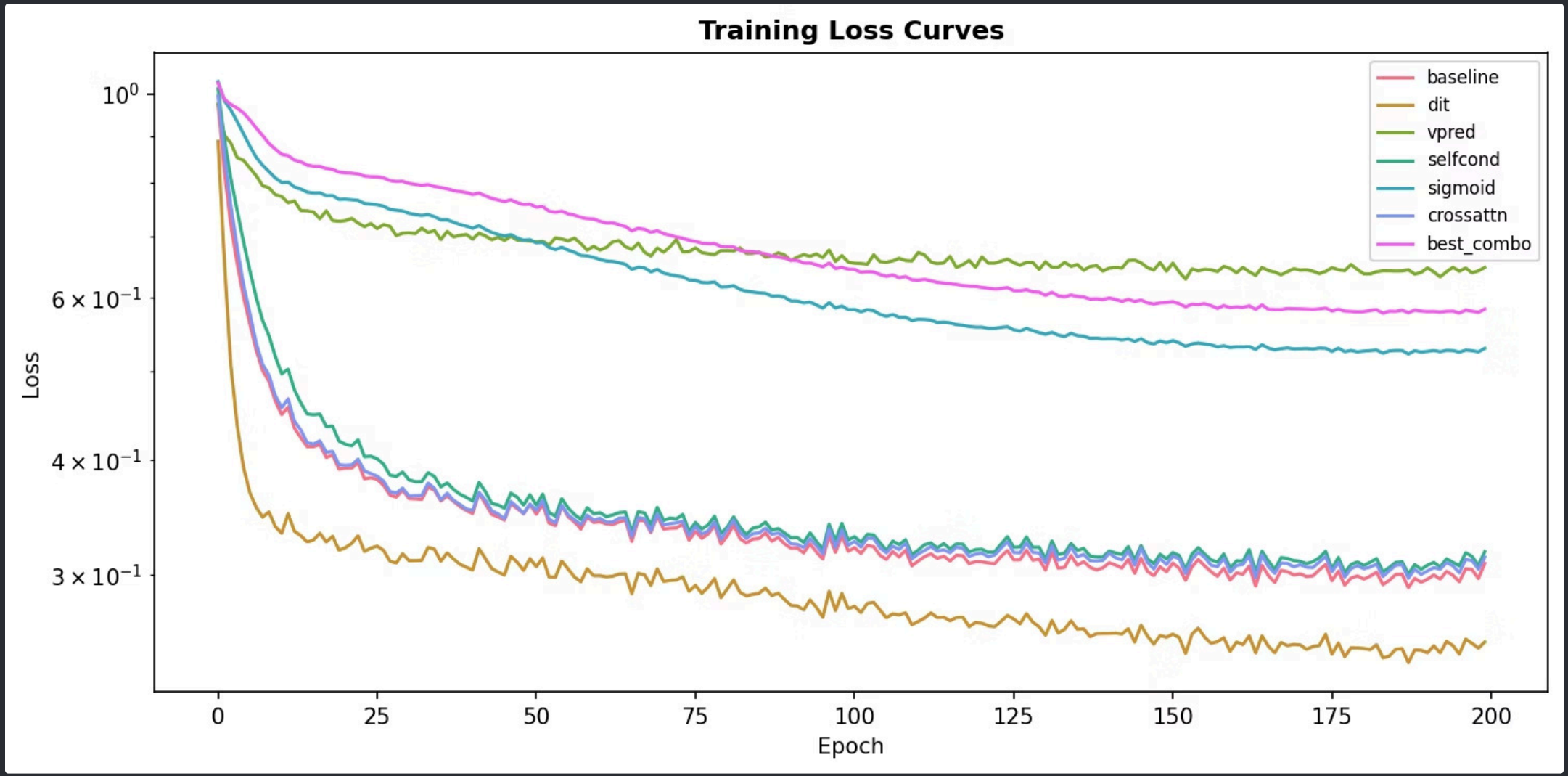
Fat tails, clustering, leverage, long memory, cross-asset ✓

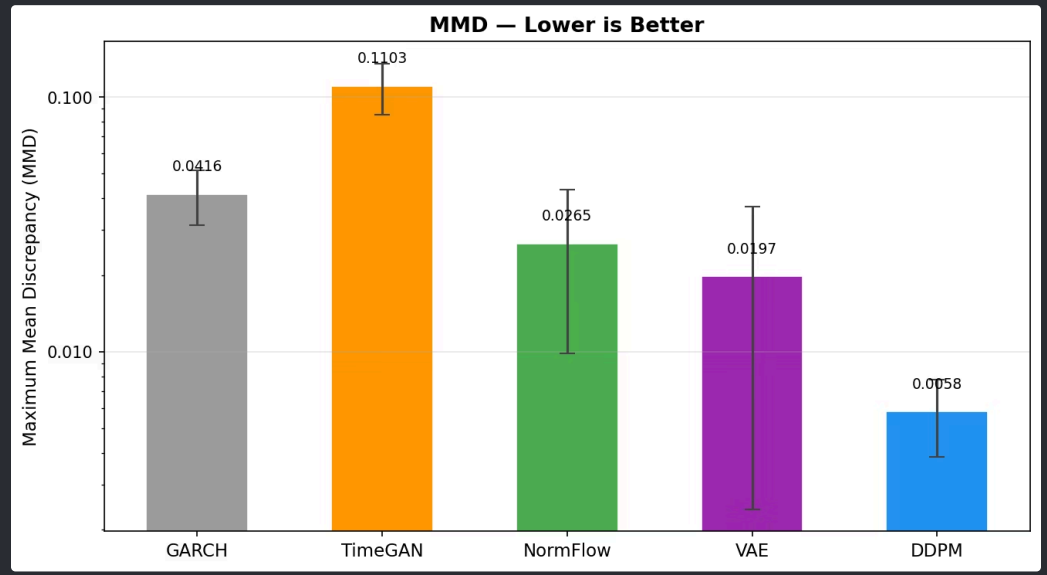
MMD: 0.006

Best by far — 4.5× better than RealNVP

Disc: 0.85

Highly realistic synthetic paths

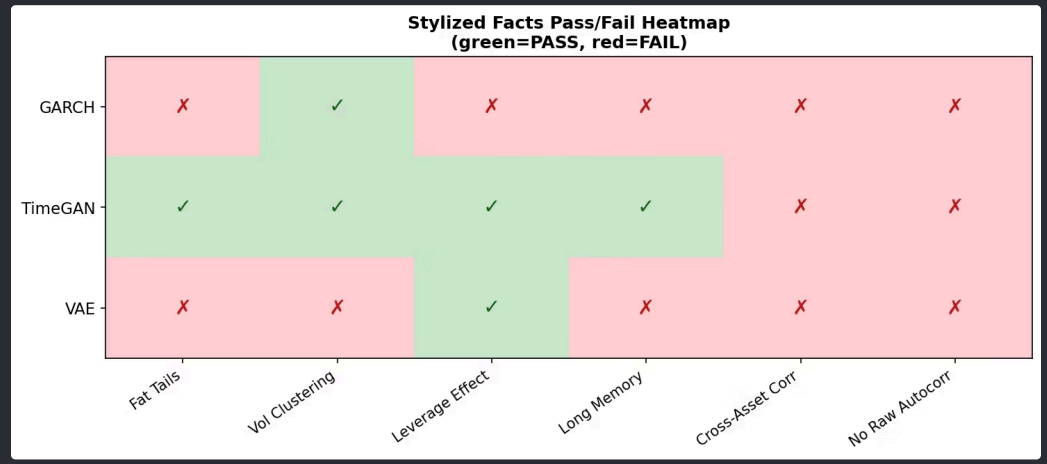

Training Loss Curves
baseline
dit
vpred
selfcond
sigmoid
crossattn
best_combo
Loss
10^0
6×10^{-1}
4×10^{-1}
3×10^{-1}
0
25
50
75
100
125
150
175
200
Epoch



Distributional Fidelity (MMD)

Maximum Mean Discrepancy — Lower is Better

DDPM achieves the lowest MMD of 0.006 — nearly 4.5× better than NormFlow and 18× better than TimeGAN. MMD measures how closely the synthetic return distribution matches the real one; DDPM's Student-t forward process is the key driver of this gap.



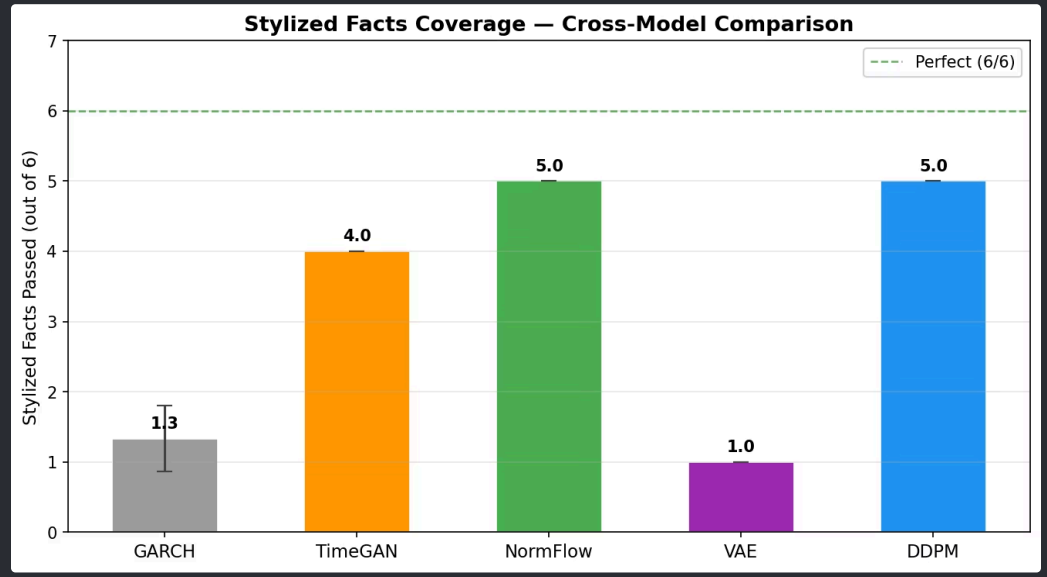
Stylized Facts Heatmap

Per-Fact Pass/Fail Breakdown (Baseline Models)

Among the baseline and weaker models, TimeGAN is the strongest — passing Fat Tails, Vol Clustering, Leverage Effect, and Long Memory. GARCH only passes Vol Clustering, while VAE passes Leverage Effect alone. Notably, No Raw Autocorrelation remains the hardest fact for all three models to satisfy.

Cross-Model Comparison

Model	SF Passed	MMD ↓	Disc. Score	Notes
GARCH (Baseline)	1.3 / 6	0.042	1.00	Classical reference
VAE (Improved)	1.0 / 6	0.020	0.75	Low MMD, poor SF
TimeGAN	4.0 / 6	0.110	1.00	Good temporal, poor quality
NormFlow (RealNVP)	5.0 / 6	0.027	0.73	Best discriminative score
DDPM (v-pred + Student-t)	**5.0 / 6**	**0.006**	**0.85**	**Best overall fidelity**



Stylized Facts Coverage

Cross-Model Comparison

NormFlow and DDPM lead all models with 5.0/6 stylized facts passed, far ahead of GARCH (1.3) and VAE (1.0). TimeGAN captures temporal structure (4.0/6) but at the cost of distributional quality. No model yet achieves perfect 6/6 coverage.

DDPM Phase 7 Ablation Study

Ablation Results

Config	SF	MMD	Disc
Phase 6 (v-pred + Student-t)	**5/6**	**0.006**	**0.85**
+ Min-SNR + warmup (Yuxia)	5/6	0.031	0.92
+ Min-SNR + decorr_reg (Yixuan)	5/6	0.015	0.87
+ Patch stride=2 (Yizheng)	4/6	0.021	0.72

Key Findings

- All DDPM variants reach SF=5/6 (except patch variant which loses SF1)
- Phase 6 remains optimal on MMD — Min-SNR improves discriminative score but trades sample quality
- Decorrelation regularizer reduces MAA by 17% and Ljung-Box statistic by 24–39%, but cannot flip the binary SF6 result

Reserve space for: fig_ddpm_ablation_sf.png · fig_ddpm_ablation_mmd.png

⚠ CRITICAL FINDING

Evaluation Framework Calibration Discovery

Running our evaluation framework on the **real training data** (S&P 16 assets, 2000–2024, n = 318,060 observations) reveals that **real data itself passes only 3/6 stylized facts**.

SF1 — Fat Tails: FAIL

Hill α = 7.83 — large sample size yields statistically thinner measured tails than the threshold requires

SF4 — Long Memory: FAIL

Hurst = 1.01 — exceeds the 0.5–1.0 range due to non-stationarity artifacts at this scale

SF6 — No Raw Autocorrelation: FAIL

LB statistic = 5,927 — Ljung-Box is statistically impossible to pass at n=318K across 20 lags

✓ Implication: Synthetic DDPM data at 5/6 is MORE stylized-fact-compliant than the actual source data (3/6). SF=5/6 represents the effective empirical ceiling — pushing to 6/6 would be counterproductive.

Diagnostic Visualizations

Return Distributions

DDPM closely matches heavy-tailed real returns; GARCH is too Gaussian; VAE shows mode collapse.

→ *fig_distributions_h2h.png · fig_qq_h2h.png*

ACF of |Returns|

DDPM captures the slow decay characteristic of volatility clustering; GARCH decays too fast; TimeGAN exhibits artifacts.

→ *fig_acf_h2h.png*

Correlation Structure

DDPM preserves cross-asset patterns — CorrDist = 1.79, lowest among all models.

→ *fig_corr_ddpm.png*

Synthetic Price Paths

DDPM generates realistic multi-asset trajectories with proper regime behavior and volatility bursts.

→ *fig_paths_ddpm_final.png*

Conclusion & Future Work

Conclusions

- DDPM with v-prediction + Student-t noise achieves **5/6 stylized facts** and **MMD = 0.006** — best overall model
- V-prediction is the single most impactful innovation: SF 1.7/6 → 5.0/6
- Student-t noise delivers 6× further MMD reduction (0.037 → 0.006)
- SF = 5/6 is the **empirical ceiling** — real data scores only 3/6 on the same framework
- Evaluation calibration finding is an independent methodological contribution

Future Directions

- Fix Ljung-Box multiple-testing issue — apply joint test or Bonferroni correction
- Explore DDIM η sweep for optimal sampling efficiency
- Conditional generation for regime-specific crisis stress testing
- Extend to higher-frequency or longer-horizon data
- Integrate synthetic engine with downstream portfolio risk management applications

Team Contributions

Member	Contributions
Shufeng Chen	DDPM Baseline and Improved (v-prediction, Student-t); Ablation Study (7 phases); Integration, Demo, Cross-Model Comparison Pipeline
Yixuan Ye	TimeGAN; Evaluation Framework (Stylized Facts); Data Pipeline; Decorrelation Regularizer; Calibration Discovery
Yizheng Lin	VAE (Improved + Original); Data Pipeline; FRED Integration; Patch-Based Training Experiments
Kevin Sun	GARCH Baseline; Visualization Utilities
Yuxia Meng	Normalizing Flow (RealNVP); Cross-Model Analysis; DDPM Training Enhancements (Min-SNR, warmup LR)